

DATA 308: Homework 5

Multiple linear regression

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```
library(ISLR2)
library(janitor)
library(tidyverse)

data("Carseats")
```

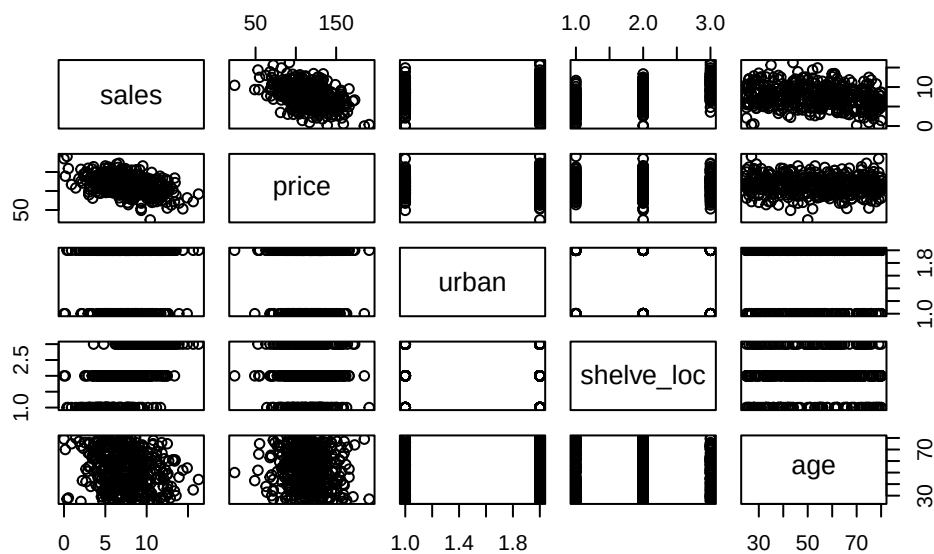
Question 1

This question should be answered using the `car_seats` data set. `car_seats` is a cleaned version of the data I made, to view the data documentation you must search `?Carseats`

```
car_seats <- Carseats |>
  mutate(
    ShelfLoc = factor(ShelveLoc, levels = c("Bad", "Medium", "Good")),
    US = factor(US, levels = c(FALSE, TRUE))
  ) |>
  clean_names()
```

Run the code below to view 2-D scatter plots of sales plotted against selected predictors.

```
car_seats |>
  select(sales, price, urban, shelve_loc, age) |>
  pairs()
```



Question 1a

Fit a multiple regression model to predict `sales` using `price`, `urban`, `shelve_loc` and `age`.

Response

```
car_seat_mlr <- lm(sales ~ price + urban + shelve_loc + age, car_seats)
summary(car_seat_mlr)
```

Call:

```
lm(formula = sales ~ price + urban + shelve_loc + age, data = car_seats)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.7390	-1.1662	0.0192	1.1601	5.1039

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.790575	0.570992	25.903	<2e-16 ***

```

price            -0.060588    0.003700 -16.376    <2e-16 ***
urbanYes         0.359849    0.191329   1.881    0.0607 .
shelve_locMedium 2.002595    0.213466   9.381    <2e-16 ***
shelve_locGood   4.974159    0.259650  19.157    <2e-16 ***
age              -0.050366    0.005405  -9.319    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.736 on 394 degrees of freedom
Multiple R-squared:  0.627, Adjusted R-squared:  0.6223
F-statistic: 132.5 on 5 and 394 DF,  p-value: < 2.2e-16

```

Question 1b

Write out the theoretical population model, being careful to handle the qualitative variables properly.

Response

$$\text{Sales} = \beta_0 + \beta_1 \times \text{Price}_i + \beta_2 \times \text{Urban}_i + \beta_3 \times \text{ShelveLoc}_i + \beta_4 \times \text{Age}_i + \epsilon_i$$

Question 1c

Write out your fitted model, being careful to handle the qualitative variables properly.

Response

$$\hat{\text{Sales}} = 14.790575 + (-0.060588 \times \text{Price}_i) + (0.359849 \times \text{Urban} = \text{Yes}_i) + (2.002595 \times \text{ShelveLoc} = \text{Medium}_i) + (4.974159 \times \text{ShelveLoc} = \text{Good}_i) - (0.050366 \times \text{Age}_i)$$

Question 1d

Provide an interpretation of each coefficient in the model. Be careful—some of the variables in the model are qualitative!

Response

For **sales**, we estimate a decrease of \$-60.59 for each unit increase in **price** , an increase in of \$359.85 for stores in **urban** locations, an increase of \$ 2002.60 on car seats with a **medium** quality shelving location in stores, an increase of \$4974.60 on car seats with a **good** quality shelving location in stores, and a decrease of \$-50.37 for every unit increase of average **age** of local population.

Question 1e

Using the original full model, obtain 95 % confidence intervals for each of the coefficients (i.e., slopes).

Response

```
confint(car_seat_mlr)
```

	2.5 %	97.5 %
(Intercept)	13.66800360	15.91314707
price	-0.06786160	-0.05331375
urbanYes	-0.01630394	0.73600216
shelve_locMedium	1.58292044	2.42226961
shelve_locGood	4.46368643	5.48463105
age	-0.06099163	-0.03973973

Question 1f

If we wanted confidence intervals with a confidence level of 99%, would they be narrower or wider than the 95% confidence intervals?

Response

Question 1g

Using the original full model, compute a 95% prediction interval for the number of sales when a car seat price is \$100, its sold in an urban location, has a good shelving location, and the average age of the local population is 30.

Response

Question 1h

We would expect price to be very related to the number of sales made. Conduct a hypothesis test to assess if any of the remaining explanatory variables are linearly related to number of car seats sold, when price is in the model.

Response

Question 1i

In which of the models (a) or (subsetting model from part h) is the standard error for the price slope larger? Why do you think that is?

Response

Question 2

In this problem we will investigate the t-statistic for the null hypothesis $H_0 : \beta = 0$ in simple linear regression without an intercept. To begin, run this code to simulate an explanatory x and a response y as follows.

```
set.seed(1) # Setting a random seed makes it reproducible. It is useful when you are going to
x <- rnorm(100) # Generated 100 standard normal random variables
y <- 2 * x + rnorm(100) # Calculates Y with more added randomness

simulated_data <- data.frame(x = x, y = y)
```

Question 2a

Perform a simple linear regression of y onto x from `simulated_data`, without an intercept. Report:

- the coefficient estimate $\hat{\beta}_1$,
- the standard error of this coefficient estimate, and
- the t-statistic and p-value associated with the null hypothesis $H_0 : \beta_1 = 0$.

(You can perform regression without an intercept using the command `lm(y ~ 0 + x, data = simulated_data)`.)

Response

Question 2b

Now perform a simple linear regression of x onto y without an intercept, and report:

- coefficient estimate $\hat{\beta}_1$,
- its standard error, and
- the corresponding t-statistic and p-values associated with the null hypothesis $H_0 : \beta_1 = 0$.

Response

Question 2c

What is the relationship between the results obtained in (2a) and (2b)? Specifically, what is the same and what is different, if anything?

Response